

LAB 3: Hadoop – Wordcount

Implement Wordcount program on Hadoop framework

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hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ jps
5809 SecondaryNameNode
5562 DataNode
5420 NameNode
6381 NodeManager
6846 Jps
6207 ResourceManager
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ mkdir wc
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cd wc
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ nano WCMapper.java
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ nano WCReducer.java
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ nano WCDriver.java
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ javac -classpath $(hadoop classpath) -d classes *.java
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ jar -cvf wc.jar -C classes/ .
added manifest
adding: WCMapper.class(in = 1759) (out= 706)(deflated 59%)
adding: WCReducer.class(in = 1541) (out= 614)(deflated 60%)
adding: WCDriver.class(in = 1769) (out= 887)(deflated 49%)
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ nano input.txt
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ hdfs dfs -rm -r /input
rm: '/input': No such file or directory
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ hdfs dfs -rm -r /output
Deleted /output
```

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hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ hdfs dfs -put input.txt /input
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ hadoop jar wc.jar WCDriver /input /output
2026-05-04 13:35:50,398 INFO client.DefaultNoHARMFalloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-04 13:35:50,530 INFO client.DefaultNoHARMFalloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-04 13:35:50,681 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool
2026-05-04 13:35:50,701 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.
2026-05-04 13:35:50,857 INFO mapred.FileInputFormat: Total input files to process : 1
2026-05-04 13:35:50,942 INFO mapreduce.JobSubmitter: number of splits:2
2026-05-04 13:35:51,034 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777881320455_0001
2026-05-04 13:35:51,034 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-04 13:35:51,116 INFO conf.Configuration: resource-types.xml not found
2026-05-04 13:35:51,116 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-05-04 13:35:51,243 INFO impl.YarnClientImpl: Submitted application application_1777881320455_0001
2026-05-04 13:35:51,258 INFO mapreduce.Job: The url to track the job: http://bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:8088/pr
2026-05-04 13:35:51,259 INFO mapreduce.Job: Running job: job_1777881320455_0001
2026-05-04 13:35:55,304 INFO mapreduce.Job: Job job_1777881320455_0001 running in uber mode : false
2026-05-04 13:35:55,306 INFO mapreduce.Job: map 0% reduce 0%
2026-05-04 13:35:59,350 INFO mapreduce.Job: map 100% reduce 0%
2026-05-04 13:36:03,371 INFO mapreduce.Job: map 100% reduce 100%
2026-05-04 13:36:03,385 INFO mapreduce.Job: Job job_1777881320455_0001 completed successfully
2026-05-04 13:36:03,444 INFO mapreduce.Job: Counters: 54
File System Counters
FILE: Number of bytes read=678
FILE: Number of bytes written=830807
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bytes written=70
0
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ hdfs dfs -cat /output/part-00000
15
a 2
b 2
e 3
f 1
h 7
i 7
j 1
l 1
m 1
o 12
r 8
s 6
t 2
u 5
w 5
y 6
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$
```